Kartikeya Bomb

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Education

Rhodes College Memphis, TN

B.S. in Computer Science, Mathematics, and Economics; GPA: 3.71/4.0 *Aug. 2023 - May 2027*

- **Coursework:** Data Structures & Algorithms, Operating Systems, Linear Algebra, Econometrics, Differential Equations, and Multivariable Calculus.
- **Activities:** CS 141 Teaching Assistant; Computer Science, Engineering, and Chess clubs; Curb Institute of Music performer.

Technical Skills

Languages: C++, C, Python, SQL, Java, TypeScript, JavaScript

AI Engineering and ML: OpenAI API, PyTorch, Hugging Face, LangChain, LangGraph, OpenCV, ONNX, LangSmith, Claude Code

Frameworks: React, Next.js, Spring Boot, FastAPI, Flask

Data and Infrastructure: PostgreSQL, Redis, Prisma, Kafka, Apache Spark, Docker, Kubernetes, AWS, GCP, Git

Experience

Enaiblr - Software Engineering Intern June 2026 - Present -

- Developed an XGBoost model with SHAP explanations to predict MATA bus failures up to 7 days in advance with 92% precision, helping technicians prioritize inspections.
- Built a Gemini and RAG workflow for the Memphis Orpheum marketing, reducing campaign drafting and review time by 60%.

AutoZone - Software Engineering Intern May 2025 - Aug. 2026 - Memphis, TN

- Built an internal React platform for AI-driven store onboarding, using Spring Boot APIs to validate bulk imports and review proposed operational changes.
- Engineered a Python AI agent that reconciled HR, store, and Bringg data, diagnosed inconsistencies, and executed remediation workflows through structured tool calls.
- Built agent state and recovery logic so AI workflows could verify completed actions, retry failed steps, and avoid repeating successful operations.
- Built a distributed pin-pad management system across 7,000+ stores using C/C++ for device communication, IP-based discovery, WebSockets, and React **saving 840 hours and $133K** .

RK Heart and Vascular Care - Software Engineering Intern Aug. 2024 - Present - Memphis, TN

- Integrated PracticeSuite with custom AI revenue-cycle platform to automate coding, claim submission, denial management, and payment posting across **300+ monthly claims**.
- Built REST/FHIR workflows with claim-state orchestration and audit logging, reducing manual billing work by **100%**.
- Integrated a rideshare system with Python/Flask and Baptist Hospital APIs, automating **250+ monthly ride requests** and cutting patient no-shows by **40%**.

Projects

LynxLifts - AI Campus Rideshare Platform | *React Native, PyTorch, PostgreSQL/PostGIS, Redis, Kafka, AWS*

- Scaled a campus rideshare platform to **200+ verified students and 800+ rides** with real-time driver tracking, messaging, payments, and push notifications.
- Built a PyTorch AI dispatch engine that ranked driver-passenger matches using route overlap, pickup ETA, schedule compatibility, and historical ride behavior, reducing median wait time by **34%**.
- Trained spatiotemporal demand models to predict ride hotspots and served real-time predictions through FastAPI, Redis, and PostGIS with sub-**150 ms** matching latency.

InnerJ - Prompt Engineering Forum | *Next.js, TypeScript, PostgreSQL, Prisma, Redis, OpenAI API*

- Built and grew a full-stack platform where **100+ users** discover, test, and share reusable AI prompts.
- Reduced feed response time by **58%** by optimizing PostgreSQL retrieval and adding Redis caching.
- Processed **1,000+ prompts** with OpenAI to automatically organize content and improve prompt discovery.

MarketPulse - Real-Time Market Data Engine | *C++20, SQL, WebSockets, SIMD, CMake, GoogleTest*

- Built a multithreaded C++ database for live cryptocurrency trade ingestion and price/volume analysis, sustaining **100,000 events/sec** in replay benchmarks on an **8-core CPU**.
- Implemented B+ tree indexes, a concurrent query worker pool, and SIMD-accelerated filtering, achieving **sub-20 ms p95 latency** for selective time-range queries across **10 million trade records**.
- Developed write-ahead logging, batched commits, and checkpoint recovery, preserving all acknowledged writes across **1,000 automated crash-and-restart tests**.

ReefWatch - Coral-Reef Monitoring Network | *Python, PyTorch, ONNX, Go, MQTT, Docker*

- Built an edge monitoring network to detect coral-reef activity in low-connectivity environments using quantized audio-vision models.
- Cut uplink traffic by **88%** with selective synchronization and added automatic rollback for reliable remote deployments.